

The Creative Media Production & Equipment Hub

Name	ID	Section Number
Hanya Essam Eldin Mohamed	20246122	G7
Habiba Amr Mohamed	20246042	G8
Marina Emad Shawky	20246087	G8
Nada Abdelhaliem	20246114	G7
Radwa Hussein Mohamed	20246050	G7
Jumana Hany Mohamed	20246039	G7

TA<Eng. Nehal Akram>

Project ID: 305

Requirements:

Supervisor:

- The system must ensure that every supervisor has a unique supervisor ID, a full name, a responsibility description, and an employee ID, so that their role and identity within the organization are always clearly defined.
- The system must allow a supervisor to register zero or more professionals into the system, because supervisors are responsible for onboarding the talent they work with.
- The system must allow a supervisor to manage zero or more studios, meaning they control and update the availability status of each studio they oversee.

Professionals:

- The system must ensure that every professional has a unique professional ID, a full name, an email address, and a role such as Director, Editor, or Cinematographer stored as a single value on the professional record.
- The system must ensure that every professional is registered by exactly one supervisor, because responsibility for onboarding must always be traceable to one accountable person.
- The system must allow a professional to execute zero or more projects, reflecting the reality that experienced professionals often contribute to multiple productions simultaneously or over time.
- The system must allow a professional to be involved in zero or more sessions, with a set date recorded for each involvement, so that attendance and participation history is always documented.
- The system may allow a professional to be part of a three-way equipment assignment linking them to a specific session and a specific

project, so that equipment responsibility is always tied to a person, a time slot, and a production.

Equipment:

- The system must ensure that every piece of equipment has a unique serial number, a condition description, a responsibility note, a category, and a model name, so that the team always knows the state and classification of every asset they own.
- The system must allow equipment to be used in one or more sessions, because the same camera, microphone, or lighting rig may be deployed across multiple productions at different times.

Project:

- The system must ensure that every project has a unique title, a total budget, a deadline date, and a current status, so that management always has a clear financial and timeline picture of ongoing productions.
- The system must ensure that every project is linked to zero or more booking, because a project cannot be initiated without a confirmed client reservation backing it.
- The system must allow a project to be executed by one or more professionals, because productions require teams of specialists working together across different roles.
- The system may allow a project to be part of a three-way equipment assignment with a professional and a session, so that it is always recorded which professional handled which equipment under which production..

Studio:

- The system must ensure that every studio has a unique studio ID, a sound recording type, a green screen indicator, a studio type, and an availability status, so that booking coordinators can immediately identify which spaces are ready for use.

- The system must allow a studio to be managed by one or more supervisors, because large production facilities may require shared administrative responsibility.
- The system must allow a studio to take place zero or more sessions over time, since studios are reused repeatedly across different productions and bookings.
- The system must ensure that every studio owns exactly one wing or more, so that every physical space is always located within a defined section of the production facility.

Session:

- The system must ensure that every session has a unique session ID, a booking reference, a date, a start time, an end time, a calculated duration, and a current status, so that the scheduling system always has complete time and resource information.
- The system must ensure that every session is reserved to exactly one booking, because a session cannot be scheduled without an approved reservation authorizing it.
- The system must allow a session to take place in one or more studios across different occasions, and must ensure that each studio hosting a session is properly recorded through the takes place relationship.
- The system must allow a session to have zero or more pieces of equipment used in it, so that the right tools are always tracked and available for the work being done.
- The system must allow a session to involve one or more professionals, with the specific date of involvement recorded, so that participation in each session is formally documented.
- The system may allow a session to be part of a three-way equipment assignment with a professional and a project, so that equipment usage is always linked to the person responsible, the session it occurred in, and the project it served.

Client:

- The system must ensure that every client has a unique client ID, a full name, a company name, a contact person name, an email address, and a phone number, so that the production team can always identify and reach the right organization.
- The system must allow a client to place one or more bookings, because a corporate client may commission multiple productions overtime .

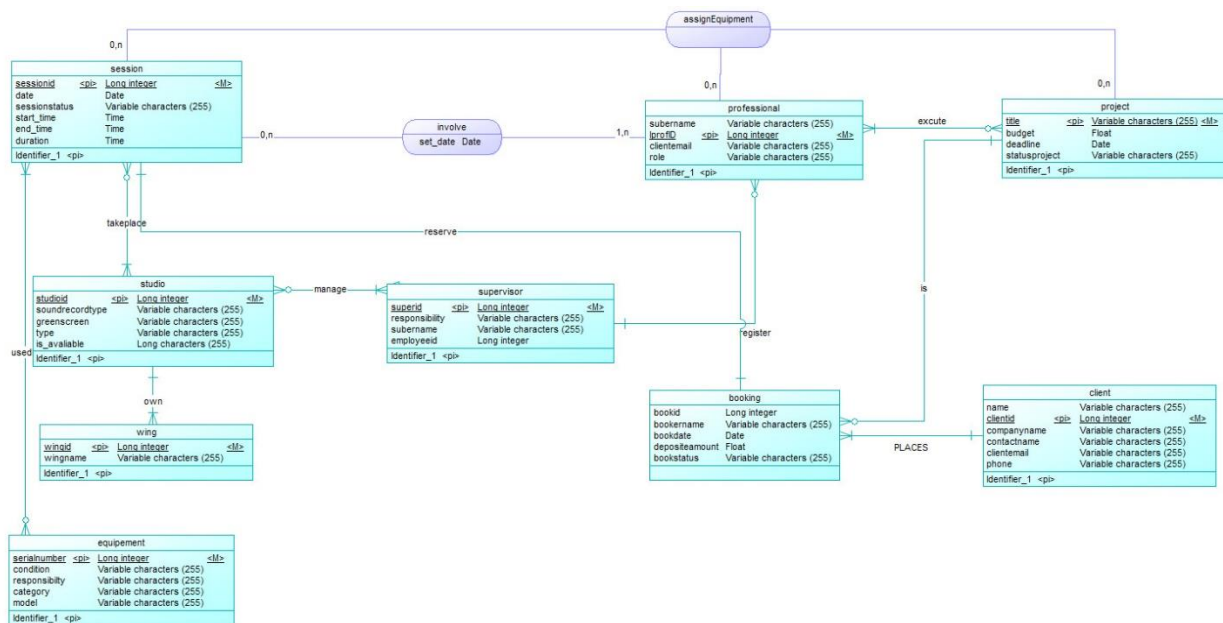
Booking:

- The system must ensure that every booking holds a unique booking ID, the name of the person who made the booking, the date the booking was made, the deposit amount paid, and the current booking status, so that all financial and administrative records remain complete and traceable.
- The system must ensure that every booking is tied to exactly one client, because a booking cannot exist without a known requesting party.
- The system must ensure that every project is linked back to exactly one booking, so that every production effort is always traceable to a confirmed reservation.
- The system must ensure that every session is reserved to exactly one booking, so that no session is ever scheduled without a formal agreement authorizing it.

Wing:

- The system must ensure that every wing has a unique wing ID and a wing name, so that different sections of the production facility can be clearly identified and referenced.
- The system must allow a wing to be owned by one studio, reflecting the physical structure of the facility.

Conceptual Data Model:



Physical Data Model:

